

TWENTY NINE PALMS, CA, USA

WMO: **690150**Lat: **34.300N**Lon: **116.167W**Elev: **625**StdP: **94.04**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **93121**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.6	1.2	-20.7	0.6	10.9	-18.5	0.8	10.4	15.1	9.5	12.9	10.1	1.9	310	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	42.4	20.1	41.0	19.6	39.6	18.9	23.4	36.5	22.4	35.2	21.6	34.6	4.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	15.9	28.3	18.4	14.3	29.3	16.9	13.0	30.8	72.7	36.5	68.6	34.9	65.5	34.5	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	10.9	9.1	DB	-4.1	44.8	2.3	2.0	-5.7	46.3	-7.1	47.4	-8.3	48.5	-10.0	50.0	(4)
(5)			WB	-7.1	23.9	1.9	1.1	-8.5	24.7	-9.6	25.3	-10.7	25.9	-12.1	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	10.3	12.2	16.0	19.2	23.9	29.4	32.3	31.7	27.9	20.9	14.5	9.8
	DBStd	8.70	3.35	3.63	3.77	4.17	4.20	3.53	2.59	2.31	3.65	3.66	3.77	3.22
	HDD10.0	110	37	18	2	1	0	0	0	0	0	0	8	44
	HDD18.3	966	248	174	93	39	7	0	0	0	0	18	121	266
	CDD10.0	4022	47	78	187	276	430	581	693	674	536	338	144	37
	CDD18.3	1838	0	1	19	65	179	331	434	416	287	98	8	0
	CDH23.3	26736	2	37	272	817	2231	4940	6893	6421	3871	1109	142	1
CDH26.7	16328	0	3	61	316	1157	3148	4630	4211	2309	469	26	0	
Wind	WSAvg	3.6	3.1	3.5	4.0	4.7	4.6	4.1	3.8	3.4	3.1	3.1	3.0	3.1
Precipitation	PrecAvg	119	27	24	13	4	1	1	8	7	9	6	9	15
	PrecMax	260	164	79	63	23	8	6	47	28	81	38	40	78
	PrecMin	43	3	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	62	38	22	15	4	2	2	11	8	16	9	9	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	26.6	30.9	35.0	39.2	44.1	44.1	43.0	45.0	35.7	29.1	23.1
		MCWB	9.9	11.6	13.5	15.7	17.7	19.2	21.1	21.3	19.8	17.1	13.3	10.6
	2%	DB	21.2	24.3	28.2	32.5	36.9	41.3	42.4	41.6	39.1	33.1	27.0	20.8
		MCWB	9.1	10.9	12.5	14.6	16.4	18.3	20.5	20.2	18.8	15.7	12.1	9.8
	5%	DB	19.4	22.1	26.4	30.3	34.9	39.6	41.0	40.2	37.5	31.4	24.8	18.7
		MCWB	8.4	10.1	11.8	13.6	15.3	17.6	20.0	19.7	19.0	15.0	11.5	8.6
	10%	DB	17.4	19.8	24.5	27.9	32.9	38.0	39.5	38.9	36.2	29.3	22.3	16.6
		MCWB	7.9	9.2	10.9	12.7	14.8	17.6	19.4	19.6	18.1	14.0	10.6	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	13.8	14.3	17.3	20.2	22.8	25.5	25.9	23.1	19.1	14.8	13.0
		MCDB	17.2	22.5	27.8	31.8	37.2	39.5	38.2	38.2	36.7	29.4	25.0	19.5
	2%	WB	10.8	12.1	13.1	15.5	17.6	20.8	23.3	23.7	21.7	17.7	13.3	11.5
		MCDB	17.2	20.7	26.1	30.6	34.0	38.0	36.3	36.5	34.9	29.1	24.7	17.1
	5%	WB	9.9	11.0	12.3	14.2	16.2	19.6	22.3	22.5	20.8	16.3	12.1	10.1
		MCDB	16.8	19.4	24.9	28.5	32.3	37.3	34.9	34.8	33.4	29.2	22.9	16.4
	10%	WB	8.7	10.0	11.4	13.0	15.2	18.3	21.6	21.6	19.9	15.1	11.0	8.8
		MCDB	16.5	18.3	23.3	26.6	31.3	35.6	34.4	34.2	32.1	27.7	21.4	15.5
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.9	13.8	14.2	14.3	15.4	13.9	14.2	14.6	14.6	13.7	12.5
		MCDBR	15.2	15.9	15.9	16.5	16.0	16.3	14.7	15.3	15.8	17.0	16.6	15.4
		MCWBR	8.0	7.5	7.0	7.2	5.8	6.1	5.9	6.2	6.6	7.9	8.3	8.1
	5% WB	MCDBR	12.9	14.0	15.0	15.6	15.3	15.6	12.7	13.1	13.6	15.6	14.9	12.9
	MCWBR	7.1	7.0	6.8	7.4	6.2	7.8	6.3	6.9	6.4	7.8	7.9	7.1	
Clear-Sky Solar Irradiance	taub	0.272	0.277	0.291	0.316	0.327	0.324	0.359	0.352	0.325	0.302	0.276	0.275	
	taud	2.550	2.557	2.502	2.422	2.420	2.443	2.429	2.433	2.485	2.522	2.574	2.567	
	Ebn at Noon	953	983	988	969	955	953	918	922	937	939	941	929	
	Edh at Noon	79	88	101	115	117	114	115	112	101	90	76	74	
All-Sky Solar Radiation	RadAvg	3.33	4.32	5.79	7.13	8.01	8.50	7.68	7.18	6.30	4.93	3.69	3.00	
	RadStd	0.22	0.34	0.23	0.21	0.27	0.30	0.35	0.29	0.22	0.22	0.15	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (38 neighbors)	+0.58	N/A	-1.71	+0.58	N/A	N/A	N/A	-91	+204	+112

Nomenclature: See separate page